StarCraft 2, Bronze to Grandmaster: refit

Setting	Selected model	chi2 / df	p	AIC
raw percentiles, 8 mechanisms	skill + practice + drive	33.0 / 9	0.00013	39.0
raw percentiles, 10 mechanisms	skill + practice + drive + evidence	26.9 / 8	0.00073	34.9
lattice-aware, 8 mechanisms	skill + practice + drive + lapse	22.3 / 7	0.0022	32.3
lattice-aware, 10 mechanisms	skill + practice + drive + lapse	22.3 / 7	0.0022	32.3

Sources: 4_output/s1_refit_summary.csv; fits in 4_output/s1_lattice/ and, for raw percentiles with 8 mechanisms, 4_output/mech_selection/S1.json. AIC values compare models within one setting only: raw and lattice-aware settings fit different observations.

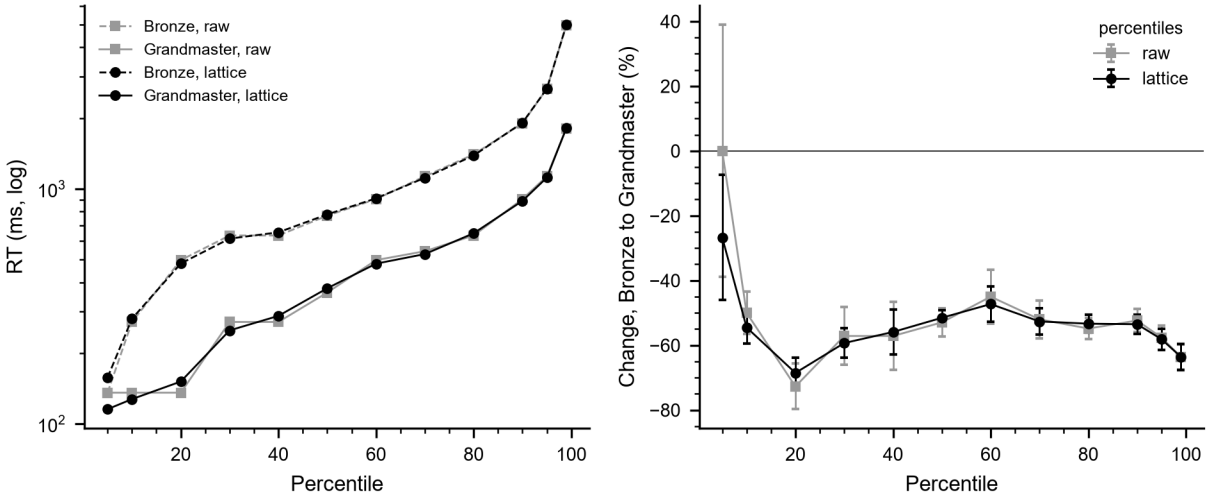


Figure 1. StarCraft 2 RTs, Bronze (167 games) and Grandmaster (32 games), pooled. Every RT is a whole multiple of 45.198 ms. Raw percentiles (grey squares) sit on those steps; lattice-aware percentiles (black circles) spread each RT evenly over its step. Right: percent change with 95% intervals (1.96 SE, cluster bootstrap over games, 500 replicates). Source: 4_output/s1_lattice_profile.csv.

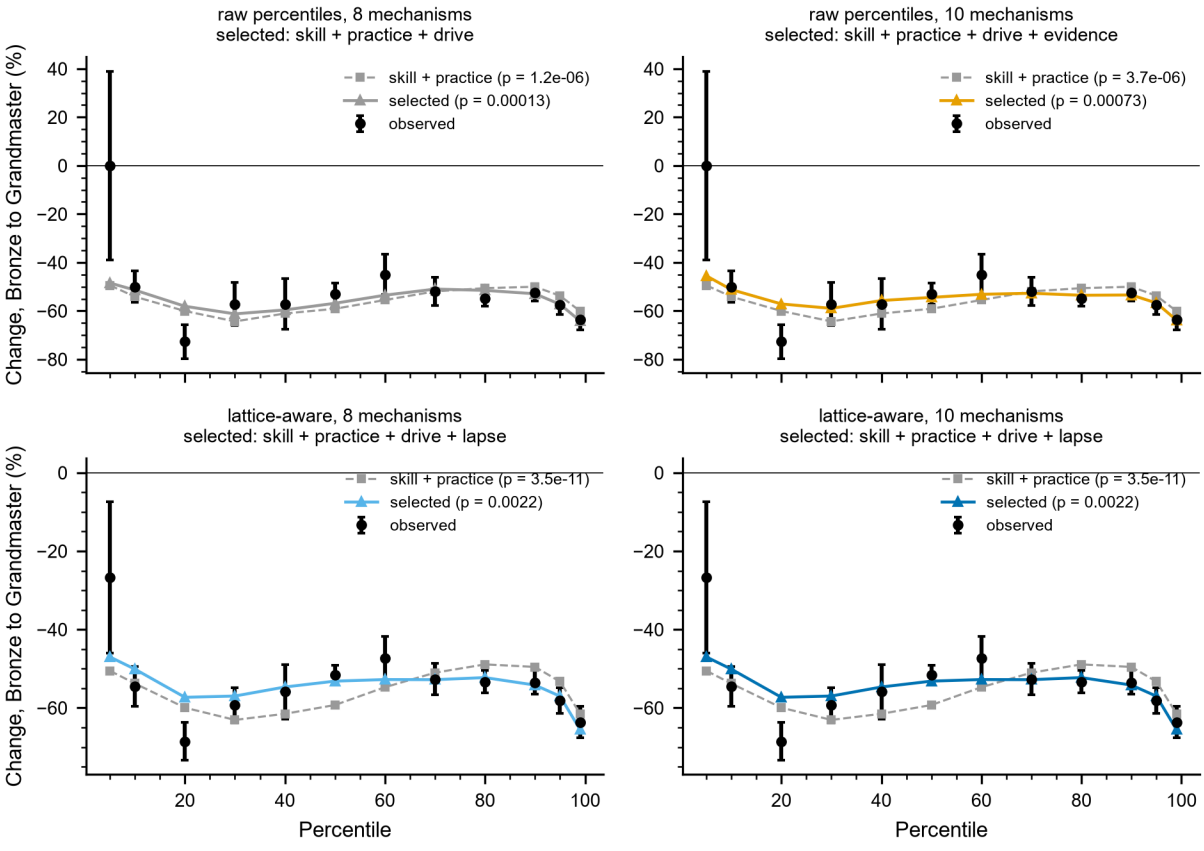


Figure 2. Observed change (95% intervals) and model predictions in the four settings: percentiles raw or lattice-aware, and stage 15's eight mechanisms or those plus circuit strength and retrieval strength. Selected model by forward selection on AIC (stage 15's rule). p from a chi-square test over the 12 percentiles with data and simulation SE; a model is adequate at $p > .05$. Raw, 8 mechanisms is stage 15's S1 fit.

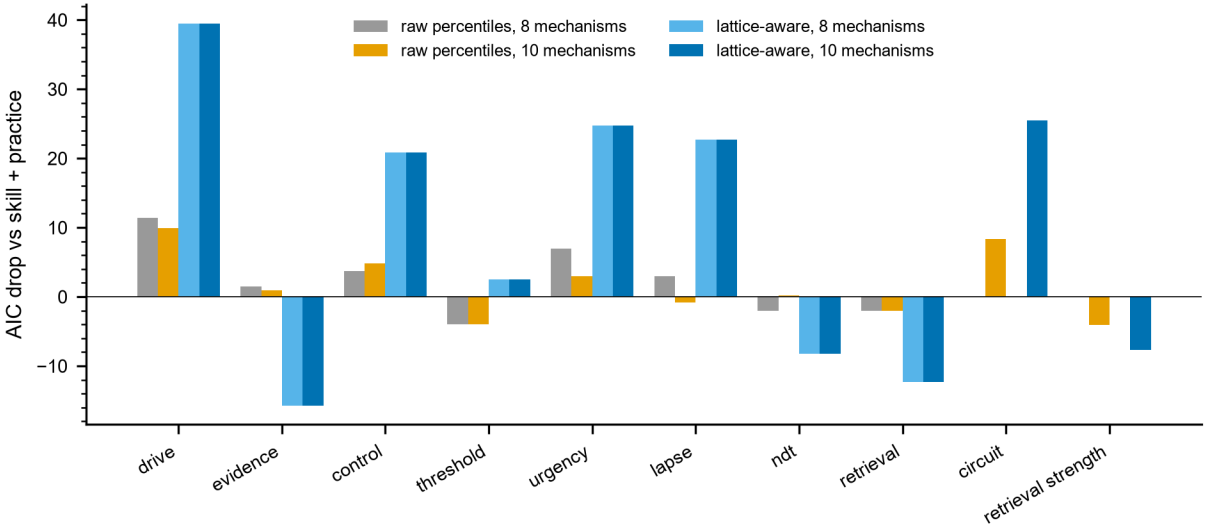


Figure 3. AIC drop when each mechanism alone is added to skill + practice (first selection step), in each setting. Larger = more of the league difference accounted for. Circuit and retrieval strength are absent from the 8-mechanism settings.